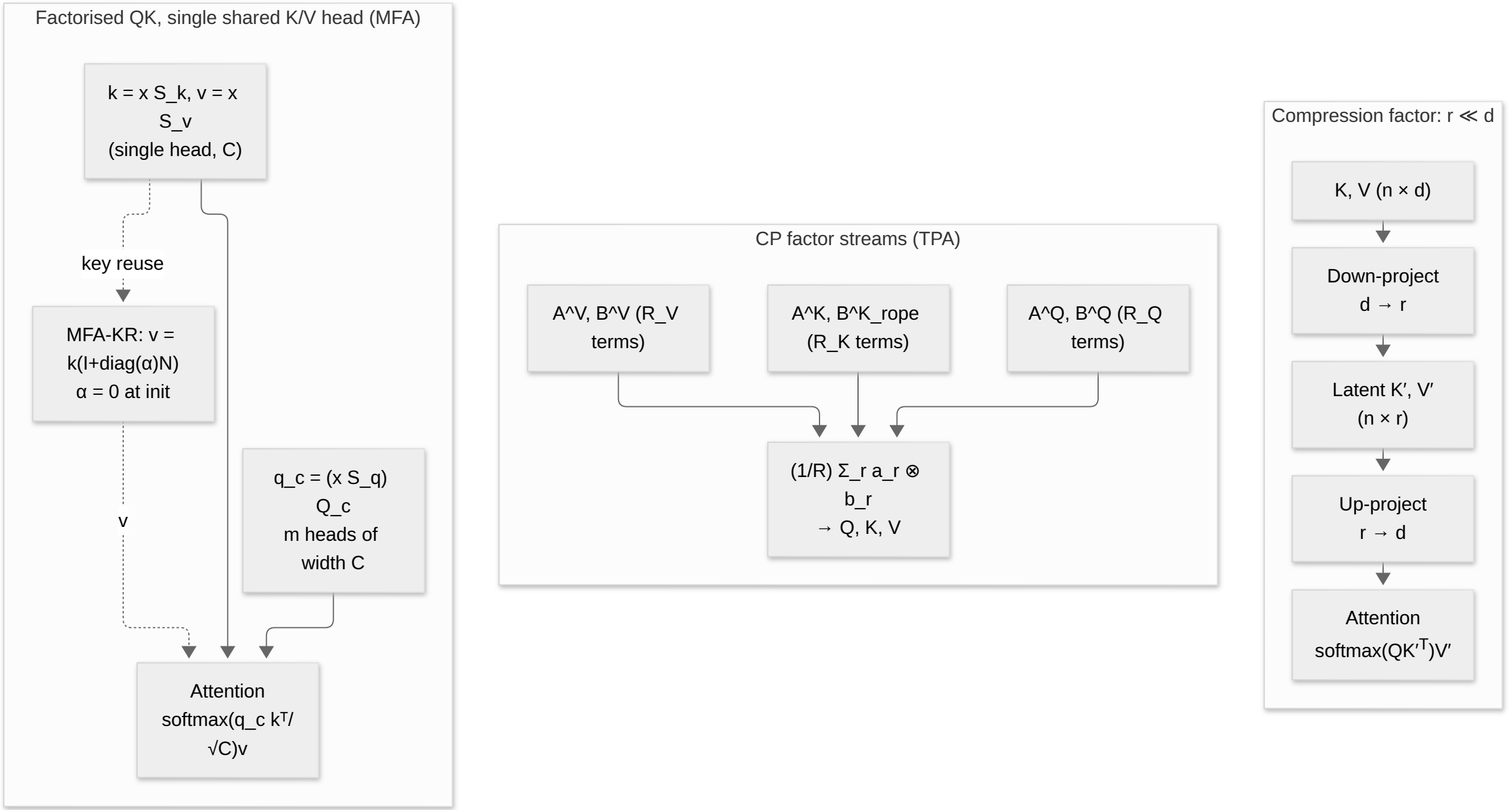

Factorised QK, single shared K/V head (MFA)
k = x S_k, v = x S_v (single head, C)
key reuse
MFA-KR: v = k(I+diag(α)N) α = 0 at init
v
q_c = (x S_q) Q_c m heads of width C
Attention softmax(q_c kᵀ/√C)v
CP factor streams (TPA)
A^V, B^V (R_V terms)
A^K, B^K_rope (R_K terms)
A^Q, B^Q (R_Q terms)
(1/R) Σ_r a_r ⊗ b_r → Q, K, V
Compression factor: r ≪ d
K, V (n × d)
Down-project d → r
Latent K′, V′ (n × r)
Up-project r → d
Attention softmax(QK′ᵀ)V′